

PROJECT PLAN DOCUMENT

Project number	28
Project Title	Fleet Management
Document	Project Plan
Creation date	January 21, 2025
Created By	Akshitha
Client	Akhil Reddy Anthamgari

Problem Statement

The topic for this project is the challenge of optimizing employee transportation by designing efficient routes in situations with many pickup and drop-offs. The problem is minimizing travel time and traveling cost by adhering to other constraints like vehicle capacities and schedules of drivers and passengers. Again, there are two of the most common cases, for instance, sending occupants from various points destined-off and a single destination point (Many-to-One), or managing lengthy commutes with multiple start and end points (Many-to-Many).

Team Members

1. **Akshitha** : Geocode employee addresses and Route Visualisation
2. **Aryanil** : Optimization Algorithm
3. **Aviral** : Integration of the algorithms and Communication
4. **Satvik** : Preprocessing
5. **Sherley** : Data Acquisition and Import

Team Communication

- Meet every Saturday at 2PM at workspace.

Development Environment:

IDE:

- Visual Studio Code

Programming Languages:

- Frontend: TypeScript (or) Javascript with React for the user interface
- Backend: Next.js or Nest.js

Open-Source Tools for Route Optimization:

- OpenStreetMap (OSM): Provides geographic data for mapping
- pgRouting: Geospatial routing extension for PostgreSQL
- Open Source Routing Machine (OSRM): High-performance routing engine for route calculations

- GraphHopper: Routing engine with support for various transportation modes
- VROOM: Designed for solving Vehicle Routing Problems (VRP)

Visualization Libraries:

- Leaflet: For displaying interactive maps
- OpenLayers: For more complex map visualizations

Collaboration Tools:

- Communication: Gmail,Whatsapp
- Documentation: Microsoft Word,Ms Excel, GitHub

Data Acquisition and Processing Tools:

- Osmosis or Overpass API: For downloading OSM data
- PostgreSQL with PostGIS: For managing geographic data
- MongoDB as a Database

Optimization Techniques and Algorithms:

- Traveling Salesperson Problem (TSP) Algorithms: For single-route optimizations
- Vehicle Routing Problem (VRP) Algorithms: For multi-vehicle optimizations
- Heuristic Algorithms: Genetic Algorithms or Simulated Annealing for complex scenarios

Additional Considerations:

Real-Time Traffic Data: Integration with Google Maps or TomTom

Milestone Schedule

Milestone	Due Date	Release	Deliverable?
Create draft requirements	23/01/25	R1	No
Finalize requirements	26/01/25	R1	Yes
Project setup and planning	02/02/25	R1	No
Initial documentation	31/01/25	R1	Yes
OpenStreetMap integration	09/02/25	R1	Yes
Road network graph implementation	16/02/25	R1	No
Geocoding system for employee locations	23/02/25	R1	No
Basic routing algorithm	01/03/25	R1	Yes
Many-to-One routing optimization	08/03/25	R1	Yes

<i>Simple user interface for data input</i>	<i>15/03/25</i>	<i>R1</i>	<i>Yes</i>
<i>Basic route visualization</i>	<i>22/03/25</i>	<i>R2</i>	<i>Yes</i>
<i>Many-to-Many routing capability</i>	<i>29/03/25</i>	<i>R2</i>	<i>Yes</i>
<i>Real-time traffic integration</i>	<i>05/04/25</i>	<i>R2</i>	<i>No</i>
<i>Dynamic rerouting functionality</i>	<i>12/04/25</i>	<i>R2</i>	<i>Yes</i>
<i>Final testing and bug fixes</i>	<i>15/04/25</i>	<i>R2</i>	<i>No</i>
<i>Documentation completion</i>	<i>19/04/25</i>	<i>R2</i>	<i>Yes</i>
<i>Final Delivery</i>	<i>26/04/25</i>	<i>R2</i>	<i>Yes</i>